

LABORATORY INFORMATION SYSTEM (LIS)

JharLab Pathology Software

Premium & Intelligent Offline Software for Diagnostic Labs (NABL Standards Compliance)

Namaste! Agar aap apne pathology lab client ko yeh software introduce kar rahe hain, toh unhe asaan aur clear Hinglish language mein features samjhana zaroori hai. Yeh list unko software ki utility, speed, aur business value samajhne mein madad karegi.

1 Core Operations & Patient Experience

Quick Patient Registration & Smart ID System

Patient ki registration details add karna bohot aasan hai. Mobile number, age, gender aur billing parameters enter karte hi system automatically unique Patient Lab ID aur barcode/QR format generate kar deta hai.

Unique Patient ID

Emergency Marking

History Tracking

Sleek Billing & GST Invoice Management

Billing screen par automatic Subtotal, Flat/Percentage Discount aur CGST/SGST calculations hoti hain. Multi-payment system hai (Cash, Card, UPI, etc.) aur payment status (Paid, Unpaid, Due) track karna bohot simple hai.

GST Calculator

Multi-payment Modes

Invoice Print

Auto-Formula Calculation (Calculated Parameters)

Kuch test results system khud calculate kar leta hai. Jaise jab technician Total Protein aur Albumin ki values daalega, toh Globulin (= Protein - Albumin) aur A/G Ratio automatically software generate kar dega bina kisi manually typing ke.

Time Saving

Zero Math Errors

Lipid & Kidney Profiles

Smart Reference Ranges & Critical Flag Warnings

Umar (Age) aur Gender (Male/Female/Other) ke base par normal ranges automatically print hoti hain. Agar patient ka result normal se bohot upar ya neeche hai, toh alert flags (↑ high, ↓ low, !! critical) dynamic display hote hain.

[NABL Standards](#)[Auto Flags](#)[Age/Gender Filter](#)

2 Automation & Machine Interfacing (LIS Interfacing)

*JharLab ki sabse badi taaqat hai iska **Machine Interfacing Engine**. Lab me moujood automated machinery ko software ke sath direct jodkar system ko auto-pilot par chalaya ja sakta hai.*

— Bi-Directional Machine Communication (LAN & COM Ports)

Sysmex, Mindray, ya kisi bhi biochemistry aur hematology analyzers ko LAN net port ya Serial (COM) port se direct integrate kar sakte hain. Bi-directional communication se machine LIS software se test order automatic read kar leti hai.

TCP/IP Server & Client

ASTM & HL7 Support

COM Port Listener

i Auto-Validation & Result Reconciliation

Jab machine sample scan karti hai aur reports ready karti hai, toh LIS result data automatically capture kar ke database me store kar deta hai. Agar koi report unmatched (Orphan) hai, toh use manual reconcile karne ki screen di gayi hai.

Zero Data Loss

Orphan Scan Reconcile

Realtime Logs

☰ Daily Quality Control (QC Result Manager)

Diagnostic accuracy banaye rakhne ke liye Quality Control entries (measured value vs expected target values, standard deviations, CV) register kar sakte hain. System batata hai ki QC Status PASS hai, WARNING hai ya FAIL hai.

QC Pass/Fail Logs

NABL Compliance

Calibration History

3 Business Suite & Operational Tools

Referral Doctor Commission Ledger

Referral doctors ki poori directory manage karein. Kis doctor ne kitne patients bheje aur unka category-wise commission payout (Biochemistry 10%, Hematology 15% etc.) kitna bana, iska ek automated ledger ban jata hai.

[Doctor Ledger](#)[Category Commissions](#)[Payout Tracker](#)

Reagent & Consumables Inventory Management

Chemicals, reagents, aur diagnostic kits ka real-time stock control. Minimum alerts thresholds ke saath expire dates track karein taaki important tests material ki shortage ki wajah se kabhi ruk na paayein.

[Low Stock Alerts](#)[Expiry Dates Manager](#)[Supplier Directory](#)

Outsource Test & Expense Tracker

Jo tests aapki lab me nahi hote aur aap unhe external path labs me outsourcing ke liye bhejte hain unka cost aur status record rakhein. Reagent purchases aur daily expenses (Salary, Bill, rent) feed karke Net Profit generate karein.

[Outsource Tracker](#)[Expense Manager](#)[Net Profit Tracker](#)

4 System Customizations & Security

Strict User Roles & Audit Trail Logs

Staff roles manage karein. Receptionist, Technician, Pathologist aur Admin ke liye limited screens ka option hai. User kab login hua aur kisne kis test key value me changes kiye, yeh poora audit trail logs me save hota hai.

[RBAC \(Security\)](#)[Activity Audit Trail](#)[Anti-Tampering](#)

100% Offline Capability with Local SQLite DB & Auto Backups

Software पूरी तरह offline काम करता है, internet band होने पर भी lab रुकेगी नहीं. Backups banana बहुत easy है और automatic recovery systems integrated हैं.

[No Internet Required](#)[SQLite Engine](#)[Automatic Backups](#)

Fully Customizable Settings (Margin, Footer & Header Printing)

Agar lab pre-printed letterhead pad use karti है, तो top margin layout को precise inches में set कर सकते हैं. Doctor और technician digital signatures, logo, stamp और footer instructions easily set kiye जा सकते हैं.

[Print Margin Settings](#)[Digital Signature Upload](#)[Custom Stamp & Logo](#)

Seamless Online Updates & Licensing Security

Internet connect होने पर admin portal से push किया गया new features update one-click में download और install हो जाता है. Cryptographic machine GUID check के साथ licenses manage hotay हैं.

[Automatic Online Updates](#)[AES-256 License Keys](#)[7-Day Free Trial](#)

5 Technology Stack Overview

JharLab LIS modern और robust technologies से बना है जो extreme safety और zero lag performance maintain karti है:

✓ **Frontend UI:** Next.js & React (Tailwind CSS based dark/light dashboard)

- ✓ **Desktop Engine:** Electron (Provides native Windows application experience)
- ✓ **Database:** SQLite with Prisma ORM (High-speed relational storage, zero configuration required)
- ✓ **Automation Port:** Node.js net sockets (ASTM / HL7 communication)